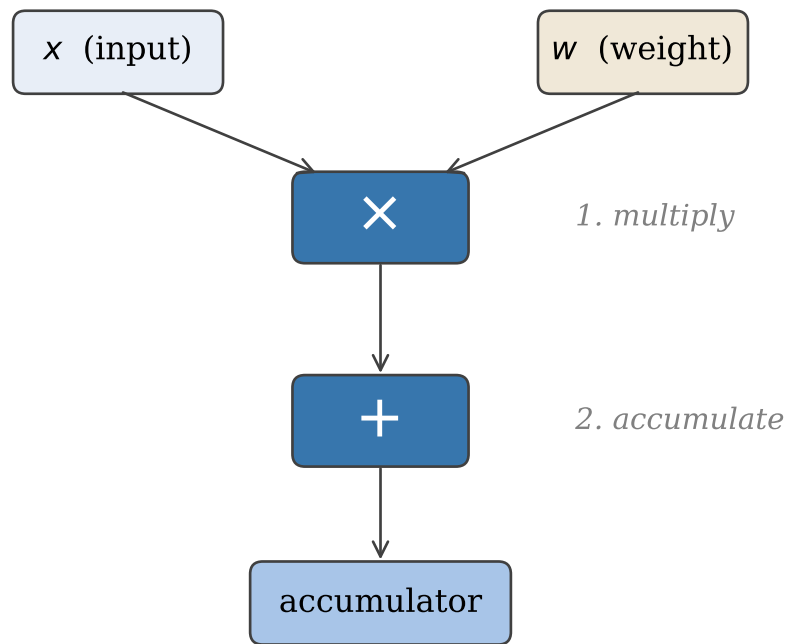


MAC (Multiply-Accumulate)

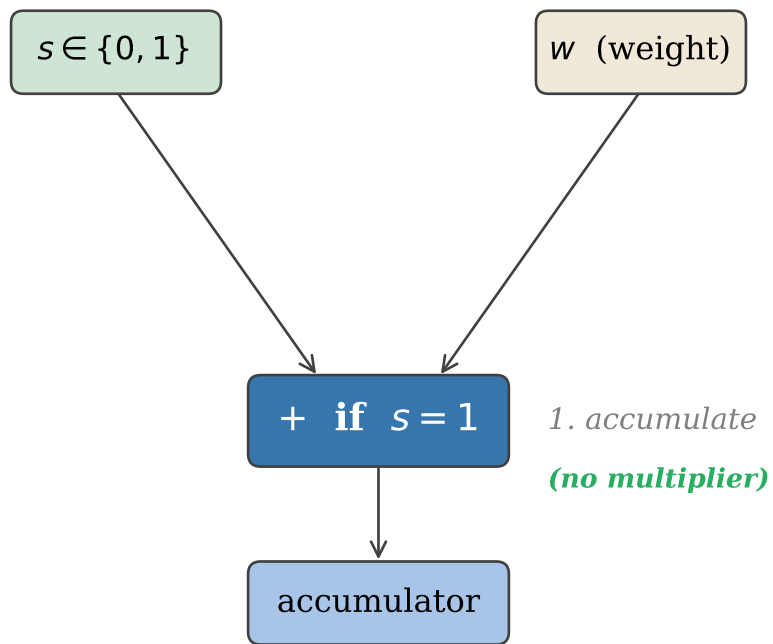
CNN: continuous floating-point



$$E_{\text{MAC}} = 4.6 \text{ pJ}$$

AC (Conditional Add)

SNN: binary spike eliminates multiplication



$$E_{\text{AC}} = 0.9 \text{ pJ}$$

5× reduction in per-operation energy at the silicon level